Professor Peter Chinedum Nnabude has been involved in Teaching and Research as well as project formulation and execution in Agriculture, Ecology and Environment since 1987. He obtained his M.Sc. and Ph.D in 1986 and 1995 respectively from the University of Nigeria, Nsukka and further training in Advanced Soil Management and Irrigation practice at the institute of Soil and Water, Bet-Dagan, Israel.

Apart from his routine teaching and research duties, Professor Nnabude has been involved in a number of Research and Agricultural/Environmental projects at local, national and international levels and has to his credit numerous publications in reputable journals and workshop papers. He has travelled extensively in connection with projects in Soil Science, Agriculture and Environment and has participated and held associateship positions at the Abdul Salam International Centre for Theoretical Physics (ICTP), Trieste, Italy; the World Academy of Science (TWAS) Italy; Institute for Agronomy (IAC), Brazil as an Associate Visiting Scientist and the Institute for Soil and Water, Bet-Dagan, Israel for Advanced Training in Irrigation and Soil Management. He served as the team leader of the Anambra State Focal NGO (CRMCI) that provided consultancy services to the World bank supported Nigeria Erosion & Watershed Management Project (NEWMAP) that tackled erosion projects in several States of the Federation. He is blessed with three children, two boys and a girl.

CADVENTURES NIGERIA(PUBLISHERS)

9 789787 805639

SOIL: THE INDISPENSABLE LINK

CRMCI/CADVENTURES

SOIL: THE INDISPENSABLE LINK

EDITED BY
PETER C. NNABUDE
AKUDO O. ONUNWA
PAUL N. OKPATA
IBEABUCHI UKO



SOIL: THE INDISPENSABLE LINK

Copyright 2023 by: Peter Chinedum Nnabude *et al.*
www.crmci.org.ng

PUBLISHED BY:
CADVENTURES NIGERIA (Publishers)
41, Igwe J.C. Menakaya Road, Umunya,
Anambra State, Nigeria
Tel 09043564018, 07033128992
E-mail:cadventuresng@gmail.com

ISBN: 978 – 978 – 795 – 169 - 9

All rights reserved.
No part of this publication may be reproduced, stored in a retrieval system or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise without the prior written permission of the copyright owner or publisher

Arrangement and printing by
Cadventures Press
41, Igwe J.C Menakaya Road,
Umunya, Anambra State
Nigeria.

CHAPTER FIVE

SOIL-CROP PRODUCTION RELATIONSHIP

Onunwa, A.O[1].; Okoli, N.A[2*]; Uko, I[2]; Ndukwe, O.O[2]. and Imo, C.M.[1]

[1]Department of Soil Science and Land Resources Management
[2]Department of Crop Science and Horticulture, FAG, UNIZIK
* Corresponding Author: *na.okoli@unizik.edu.ng*

INTRODUCTION

Soil is our life support system. Soils anchor roots, hold water and store nutrients. Soils are home to earthworms, termites and a myriad of micro-organisms that fix nitrogen and decompose organic matter. The relationship between crop and Soil science is mostly for the improvement of crop productivity and to ensure food security. Soil physical and chemical properties play a central role in crop growth, development and yield by influencing the availability of air, nutrients and water to crops (Cardone *et al.*, 2020). The way an agronomist relates with the soil will reflects on the crop growth and yield and therefore, the influence of soil even in the current trend of soilless media cannot be overemphasized. Soil is the loose surface material that covers most land. According to Wikipedia, (2017) soil is a mixture of organic matter, minerals, gases, liquids and organismsthat support life. Soil is a product of soil forming factors (topography, organisms, parent materials and climate interacting over a period of time). Crop growth depends on two important natural resources - soil and water (Oshunsanya, 2011).

Water is essential for plant life processes and supplying of soil nutrients in form of solutions for accessibility to plant roots. The soil provides the mechanical support and nutrientreservoir necessary for plant growth. Soil acts like a reservoir that holds water and nutrients that plants need to grow. Apart from the role of soil as an anchorage to crops, soil physical, chemical properties and soil biological life can boost or retard crop growth and yield.

Soil characteristics and their influence on crop growth and yield responses.

There are several key soil qualities and characteristics that are important for crop production. They include: nutrient availability (nutrient retention capacity against leaching losses which depends on soil organic carbon, Soil texture, base saturation, cation exchange capacity of soil and of clay fraction**,** and rooting conditions); oxygen availability to roots; salt toxicities and soil workability.

Many factors determine or limit plant growth and development and majority of them are soil dependent viz: sodification, salinization, topographical position, the soil organic matter and other nutrient contents and texture. Certain soil physical and chemical properties determine the suitability of soil as a growth medium for crops viz:

Effect of soil structure on crop growth and yield

Soil structure refers to the arrangement and organization of soil particles in the soil and the tendency of the individual soil particles to bind together in aggregates. Aggregation of soil

particles affect water and air transport which affects the movement of solutes and pollutants and affects biological activity, including plant growth. Soil structure is influenced by management practices like tillage, crop cultivation, application of fertilizers (organic and inorganic) and irrigation water. The synergy between these management practices influence crop growth and yield. A good soil tilth improves rooting and seedling emergence in crusted soils. The improvement in soil structure through the addition of organic matter and deep tillage management contributed to corn productivity (Meng *et al*., 2016). Soil amendment of sandy soil using a combination of Na-bentonite, humic acid, Na_2CO_3 and plant cellulose had a positive effect on soil water storage, soil enzymatic activities and maize yield (Zhou *et al.*, 2016). Lamidi *et al*., (2018) reported that the improvement in the yield of tomatoes grown in a lateritic soil was as a result of the organic manure application which buffered the soil and improved soil structure. Lamidi *et al*., (2018) also stated that yield of tomatoes from the buffered lateritic soil was as high as the yield of tomatoes raised in sandy loam soil.

Effect of soil texture on crop growth and yield

Soil texture refers to the proportion of sand, silt and clay sized particles that make up the mineral fraction of the soil. Soil textural type has a great influence on the available habitat for soil organisms. Finer soils have a greater number of small 'micro-pores' that provide habitat for microorganisms like bacteria and fungi and retain more amounts of nutrients. However, fine soils with clayey property will require greater energy for tillage than in coarse textured sandy soils. Coarse textured sandy soils are well drained with good aerationcapacity and low water holding capacity, easy to cultivate with low organic matter content. The texture of the soil will determine the quantity of organic matter and the presence of soil biological life present. An abundant supply of fresh organic materials will ensure a robust population of soil organisms and this shows the interrelationship between soil texture and soil biological life. This synergy will result in better crop growth vigour and yield. Akinyeleand Adigun (2006) argued that soil texture did not affect maize yield, however influenced phenotypic characteristics of maize. Chaves *et al*., 2013 refuted the report of Akinyele and Adigun (2006) and reported that yield variability in maize is dependent on soil texture, weather and maize management. Soil textural experiment in rice showed that rice grainyield in clay soil was 46 % higher than in sandy loam soil (Dou *et al*., 2016) and Diatta *et al*., (2018) observed 31 % seed yield in mungbean and 40% plant residue increase in silt loam soil when compared to loam sand soil. Soil texture can limit mungbean growth (Diatta *et al*., 2018. Loam and sandy-loam soil texture produced saffron with highest number of flower, stigma yield, daughter corm production and horizontal diameter when compared with saffron yield and yield parameters grown in sandy and clay soils (Cardone *et al.*, 2020).

Effect of soil bulk density on crop growth and yield

Soil bulk density is the weight of dry soil (M_{solids}) divided by the total soil volume (V_{soil}) (Al-Shammary *et al.*, 2018). Soil bulk density is an indication of soil compaction. It affects soil structure and determines the infiltration, available water capacity, and aeration, soil porosity, rooting depth/restrictions, soil microorganism activity, root proliferation and nutrient availability. Soil bulk density reflects soil health and the soil's ability to support crop growth, development and yield (Goodmann, 1999). Practices such as the use of heavy and large machinery use on wet soil, consistent tillage, limited crop rotation, burning, removal of crop residues, overgrazing and overstocking of land can lead to poor bulk density. However, the problem of soil bulk density can be ameliorated by using management practices that can improve soil structure such as addition of cover crops, crop

residues, compost, animal manures, use of minimum tillage and less use of heavy machineries and crop rotation. These practices can help to increase soil organic matter content, soil microbial activities, and reduce bulk density and soil compaction.

Research works on effect of soil bulk density on crop yield showed that growing of crops in compacted soils without soil amendment negatively affected crop growth and yield. High soil bulk density showed a negative correlation with tomato root growth while planting on ridge or mounds improved number of leaves, plant height and weight of tomato fruits in relation to zero tillage and manually cleared soil (Adekiya and Ojeniyi, 2002). The findings of Adekiya and Ojeniyi, 2002 were confirmed by the report of Adekiya *et al.,* 2009 who thatstated that tillage treatments of the integration of ploughing, harrowing and ridging improved tomato yield by 62% in Alfisol by reducing the soil bulk density when compared with manual clearing. In maize, Ebubekir *et al*., (2008) reported that compaction treatments caused by press wheel increased soil bulk density, penetration resistance and affected thedry matter yield negatively.

The effect of organic manure application on soil bulk density with a resultant positive yield on crop tomato was reported by Ddamulira *et al*., (2022) who stated that organic manure decreased soil bulk density and improved the growth and yield of tomatoes

Effect of soil porosity on crop growth and yield

Soil porosity is the fraction of the total soil volume that is taken up by the pore space (Nimmo, 2004). Pore space is that part of the bulk volume of the soil that is not occupied byeither mineral or organic matter rather occupied by air or water. Pore size affects soil processes such as water percolation and aeration and can have positive or negative effect on crop growth and yield depending on the soil porosity (Bray and Weil, 2008). Soil porosity isinversely related to bulk density. Loose, porous soils have lower bulk densities and greater porosities than tightly packed soils. Compaction decreases porosity as bulk density increases. Porosity varies depending on particle size and aggregation. The presence of large number of pore spaces in the soil is mostly influenced by the presence of biological life and the activities of microorganisms, soil texture and root growth and movement. The impact of soil porosity on crop growth and yield is seen in the application of organic materials to the soil which encouraged microbial activities and root growth. The synergy between organic matter, soil biota and root growth improved soil structure and porosity; and thus, improved yield of tomatoes (Roberts *et al.,* 2007; Atiyeh *et al*., 2002). Zucco *et al*., (2015) reported increase tomato yield and e attributed the improvement in yield to increase in soil microbial populations and soil structure as a result of the application of vermicompost.

Effect of soil air on crop growth and yield

Soil air is the gaseous phase of the soil. Soil air plays an important role in plant growth and the activity of soil organisms. Soil air is found in soil pores and there is equilibrium between water and air content within a soil. The presence of water in the soil displaces air from some of the pores, hence, the air content of a soil is inversely related to its water content. The main constituents of soil air nitrogen, oxygen and carbon iv oxide. The influence of soil air on crop growth and yield is in directly related to the activities of soil microorganisms. The composition of soil air is affected by the nature of soil, soil organic matter, soil moisture and vegetation (SARE, 2021). In poorly aerated soil, roots are deprived of oxygen and fade because they cannot breathe properly. Soil aeration can promote root growth and increase the photosynthetic rate and chlorophyll content, thus promoting plant growth and reducing

plant death rate under sodium chloride stress conditions (Yuan *et al*., 2019). Soil aeration affects root growth, ensure healthy vegetation and consequently high yield (Abuarab *et al*., 2019). Soil air is required by microorganisms for the release of the decomposition of plant nutrients which are necessary for the optimum crop growth and yield (Francisco *et al.*, 2000). The deficiency of soil air slows down the rate of microbial activity. Soil air takes partin the respiration of the roots of plants and soil microorganisms (Zou *et al.*, 2001). Poor plant root absorption of nutrient elements such as N, P, K, Ca, and Mg and low productivity was as a result of lack of soil aeration (McLaren and Cameron, 1996). Insufficient aeration on root development in carrots resulted in abnormally shaped roots carrot and poor yield on compact, poorly aerated soils in relation to carrot grown in well aerated soil (Islam *et al*., 1998; Olymbios and Schwabe, 1976). Growth parameters in potato significantly increased following aeration relative to non-aerated treatments for all fertilizer applications (Abuarab *et al*., 2019). Insufficient aeration of soil also leads to the development of clubroot disease of crucifers (Seiko *et al*., 1994) and wilt of arhar and gram, dieback of citrus and peach (Indrajit, 2023).

Effect of soil water on crop growth and yield

Soils can process and hold considerable amounts of water. They can take in water, and will keep doing so until they are saturated. Some of the water will steadily percolate through the soil and end up in the waterways and streams, but much of it will be retained, despite the influence of gravity. Retained water can be used by plants and other organisms, thereby contributing to land productivity and soil health. The soil's ability to retain water is strongly related to particle size; water molecules hold more tightly to the fine particles of a clay soil than to coarser particles of a sandy soil (*Shao, 2013*).The maximum amount of water that a given soil can retain is called field capacity, whereas a soil so dry that plants cannot liberate the remaining moisture from the soil particles is said to be at wilting point (Leeper and Uren, 1993). Available water is that which the plants can utilize from the soil within the range between field capacity and wilting point. It is the available water that affects crop growth and yield. Crop productivity depends on the quality of water supplied to a crop and also the soil on which it grows (Ramaswwamyreddy, 2021). Zhang *et al*., (2004) reported that severe soil water deficit decreases grain yield of wheat and maize. Soothar *et al*observed that different water regimes stimulated sunflower growth and yield. Okoli *et al.,* 2022 reported that water deficit at three months after planting resulted in poor proximate contents of cassava leaves during the late cropping season.

Effect of soil temperature on crop growth and yield

Temperature is described is the measure of hotness or coldness expressed in terms of Celsius or Fahrenheit. Soil temperature is influenced by climate, water content of the soil, colour of the soil, soil cover, depth in the soil profile and air and water flow within the soil. Soil temperature is an essential factor in cropping because it determines whether a plant can thrive in the environment. Soil temperature determines the time for sowing and planting of crops.

Soil temperature affects seed germination, seedling emergence, root and shoot growth. Seed germination rate increases with increase in soil temperature. However, higher soil temperature of above 35°C impairs germination in crops like maize and sorghum (Uguru, 2011). Lower soil temperature causes dormancy in seeds and predisposes seeds to soil borne pathogens. Generally, the temperature range of 25 – 30°C, is optimum for seed germination while temperature below or above the optimum range delays germination process, causing

seeds to become less vigorous and consequently suffer greater stress from the environment (Helosa *et al.,* 2019). Soil temperature range affects root growth in crops. Low soil temperature encourages white succulent roots with little branching while high soil temperature encourages the development of fine, dark brown roots with more branching habit (Ali, 2018). High soil temperature above 35ºC negatively affected the growth of roots and shoots, hampers pollination, causes poor anther dehiscence and leads to spikelet sterilityin rice (Hussain et al., 2019). Soil temperature of 37ºC negatively influenced the yield, yield components, grain filling and quality in the stage from late tillering to panicle initiation in rice (Yumiko *et al.,* 2010). Sweet pepper showed increase in plant height, leaf area, number and weight of fruits and dry matter contents when subjected to heating of 18ºC at minimum root zone in a greenhouse (Abdel-Mawgoud *et al*., 2005). Soil temperature level affects the systems and processes involved in soil nutrient absorption, plant root development and survival of soil microorganisms (Amare and Desta, 2021; Sabri *et al*., 2018). Low soil temperature reduces water and nutrient uptake by roots and transport within the plant (Pregitzer and King, 2005)**.** The report of Pregitzer and King, (2005), supports the increasing practice of using plastic mulches in cold climate locations than in warm climate zones to increase the root zone temperature (Amare and Desta, 2021). Development of plantdiseases by crops is related to soil temperature. Soil borne fungi multiply actively at low soil temperatures and attack seeds, leading to poor seedling emergence and poorly developed seedlings (Silva Neta *et al*., 2015).

Effect of soil colour on crop growth and yield

Soil colour is determined by organic matter content, drainage conditions, degree of oxidation and presence of specific minerals. Dark brown soil indicates the presence of organic matter and the fertility status of the soil. Pale brown or yellow soil indicates that organic matter and nutrient contents are low and this signifies poor fertility and there is needto amend the soil. Red soil indicates extensive weathering and oxidation of iron compounds in the soil. Variation in soil colour is as a result of the presence of organic matter content which affects other soil physical properties such as absorption of soil heat, retention of moisture, aeration, drainage and biological life (Shreeja, 2022). In other words, the colour ofthe soil indicates soil fertility status and thus, indirectly affects crop growth and yield (Sanchez-Maranon, 2011). However, the presence of organic matter in the soil encourages rich biological life, humus production, soil dark brown colour, and increase in crop productivity.

Effect of soil chemical properties on crop growth and yield

The chemical property of the soil determines to a great extent how well the crops will perform in a given soil. In crop production, before land preparation is carried out, soil samples are collected randomly to ascertain the nutrient contents in order to guide the farmers on the necessary soil amendments required in a particular environment. Soilchemical properties that influence crop growth and yield include; soil organic matter, soilpH and mineral nutrients (phosphorus, nitrogen, carbon, major cations (calcium, magnesium, sodium, potassium, etc), sulfur, trace metals and elements), cation exchange capacity, base saturation, salinity, sodium adsorption ratio, enzymes, and electrical conductivity. These properties affect processes such as nutrient cycling, biologic activity, soil *formation*, pollutant fate, and erosion. Soil chemical property can be improved using organic and inorganic fertilizers and have been reported by researchers to influence crop growth and yield positively. In ultisol of pH 5.16, Okoli *et al*., (2011) reported that the application of poultry manure and palm bunch ash at the ratio of 15:3 increased soil pH to

6.79 (slightly alkaline) and increased cassava fresh root yield to 28.67 t ha^{-1} while zero application of manure lowered soil ph after planting to 5.29 and decreased cassava fresh root yield to 19.67 t ha^{-1} in the early cropping season. The disparity in the yield of the soil amended and soil without amendment showed the importance of soil chemical properties especially soil pH which triggers the release of soil nutrients for the vigrous growth and yield of cassava. The findings of Okoli *et al.,* (2011) were affirmed by the report of Emma- Okafor *et al.,* (2016) who stated that soil amendment using 12 t ha^{-1} of poultry manure increased soil pH from 4.50 to 6.07 with an increase in the residual soil nutrient elements such as available phosphorus, calcium, magnesium, calcium and sodium which translated to 7 kg per plant of bunch weight in plantain while the control had a pH of 4.80 with poor residual soil nutrient and resultant poor bunch yield of plantain (0.47 kg). Ekpe *et al.,*(2021) reported that soil fertility plays an important role in the improvement of yield of irrigated upland rice when grown in degraded sandy soil.

Effect of soil biological life on crop growth and yield

Soil contains a wide array of taxonomic diversity of macro- and microorganism of various sizes and in different quantities such as bacteria, fungi, protozoa, nematodes, enchytraeids, insect larvae, ants, centipedes, millipedes and earthworm. Soil biological contents represent the direct and indirect influence of the living organisms inhabiting the soil. Effective soil biological life reflects soil health and the ability of the soil to support life (Khali *et al*., 2015). The microorganisms are the primary decomposers of the soil which perform the function of transforming organic matter into humus (Ibeawuchi *et al*., 2015). Soil biology maintains soil structure through the activities of burrowing and decomposing organisms that create pore spaces as they migrate through the soil and form soil aggregates caused by the secretions from microorganisms (Niu *et al*., 2020)**.** The arrangement and collection ofminerals as aggregates by soil organisms have significant effects on plant growth (Zhongyou *et al*., 2017). The development of aggregates is desirable in all types of soils because it promotes better drainage, aeration, and water storage. The presence of soilbiological life helps to improve crop growth and yield by decomposing organic matter and releasing nutrients to the crops for the growth activities and yield. Relationship between soil bacterial community and soil physiochemical properties tend to improve crop yield. Soil biota resulted in enriched soil life which increased crop yield in maize and biomass in wheat(Bender and Heijden, 2014). There is a positive correlation between plantain bunch yield and soil biological life in the plantain farm produced with poultry manure and mulch during late cropping season (Alagba *et al*., 2017). Tao *et al.,* (2017) reported 31.3% increase in maize productivity and yield by milk vetch and related the yield increase to changes in soil microbes and soil properties. Soil biological life is a useful indicator of soil quality and potential crop yield (Tao *et al*., 2018). Soil bacteria and fungi community from organic and inorganic amendment improved maize yield (Ma *et al*., 2017; Gholami *et al*., 2012).

Soil and activities that go on in the soil are effective means of diseases and pests control. Bacteria, fungi, nematodes, micro-arthropods and larger arthropods in soil help in thecontrol of certain crop pest and diseases. Bacteria regulate plant disease through pathogen antagonism in the rhizosphere. Certain fungi (Ascomycota) are natural enemy of many agricultural insect pests. Pappu et al. (2009) reported that certain soil nematodes could be used for bio-control of thrips.

Regulating Services of the Soil Ecosystem are predominantly control by soil organisms. Certain biological processes that take place in the soil ecosystem enhance hydrological properties of the soil such as drainage, infiltration and water holding capacity. Large soil

invertebrates called **ecosystem engineers** have developed the ability to move and work in the soil through a process called Bioturbation.

Role of soil as crop anchorage

Soil plays the role of an anchor to the plant by holding the stem firmly to the soil through the spread of the roots which moves downwards in search of nutrients and moisture. The roots of crops play the role of water and nutrient absorption and anchor to plants. Soil as an anchorage to crops is affected by soil texture, bulk density, aggregate size, porosity, aeration (permeability), and water holding capacity. Rooting and anchorage of crops is a function of soil physical properties which predisposes crops to lodging caused by weak stems or winds. According to Goodman and Ennos, (1999), plants lodge in weak soil compared with those growing in the strong soil. Soil as a medium for plant growth must serve as - physical support, water reservoir, oxygen supplier, and nutrient source. The amount of water, air and nutrients available for plant growth is affected by the soil physical properties (Ansa et al. 2022).

Soil quality varies and soils respond differently depending on the physical, chemical and biological properties of the soil. Healthy soils are basis for healthy crop (food) production and fertile soil results in high yield and better quality plants (International Year of Soils, 2015).

SOIL QUALITY ASSESSMENT AND SUITABILITY CLASSIFICATION

Land suitability is the fitness of certain area of land for specific use. Maintaining a healthy soil implies managing the land sustainably either for land use or land improvement, hence the need for quality assessment and suitability classification. There is the need for proper Soil quality assessment for effective land use, management and sustainable crop and animal productions. Hence, there must be an organized study to characterize, conduct suitability evaluation and soil quality assessment of soils. Deforestation and accelerated soil erosion which causes soil quality degradation are serious problems globally and especially in Nigeria. Soils which have moderate to high quality are marginally to moderately suitable for both tree and arable crops. Limitations to suitability include nutrient availability, nutrient retention, climate and soil physical characteristics (Adeyolanu et al 2017). Soils may have high quality but may not be highly sustainable for crop production except with good management. The soil properties considered during suitability classification include: texture, color, organic matter content, drainage condition of the soil, and bulk density.

In agriculture, soil and plant relationship/management affects soil quality, crop yield and soil nutrient dynamics (Mikha, et al., 2005).

CONCLUSION

Soil as described by Sposito (2008) is an open system that exchanges matter and energy with surrounding atmosphere, biosphere and hydrosphere to ensure ecosystem stability. Such exchanges are of direct and indirect benefits to the existence of man on earth. Effective discharge of these services which the soil renders to the wellbeing of man is largely driven by the diversity and activities of soil organisms (macro and micro) which play complementary roles in making nutrients available and creating suitable physical soil conditions for plant growth and development

Soil is a living, dynamic ecosystem and its influence on crop productivity is a function of the sum total of the soil physical, chemical and biological properties. The synergy between soil properties influences crop growth and yield. The presence of organic matter in the soil enhances soil biota, soil chemical properties and improves soil structure. The activities of soil organisms results in the improvement of soil physical and chemical properties which interact to improve crop growth and yield. Therefore, soil continues to play an active role asan anchor and nutrient supply medium to crops and if managed properly will continue to support crop growth, development and yield and sustain food security

A healthy soil also contributes to mitigating climate change by maintaining or increasing its carbon content. Food availability relies on soils: nutritious and good quality food and animal fodder can only be produced if our soils are healthy and living.

REFERENCES

Adeyolanu, O.D, Are, K.S, Adelana, A.O, Denton, A.O and Oluwatosin, G.A (2017) *Characterization, Suitability Evaluation and Soil Quality Assessment of Three Soils of Sedimentary Formation for Sustainable Crop Production.* Journal of Agriculture and Ecology Research International, 11 (2):1-10. ISSN 23941073 https://doi.org/10.9734/JAERI/2017/31943

Ansa J., Eeman A., Khansaa B.A., Asma O., Samreen R. (2022). Soil Fertility: Factors Affecting Soil Fertility, and Biodiversity Responsible for Soil Fertility. International Journal of Plant, Animal and Environmental Sciences 12: 21-33.

Braat, L. C and de Groot R (2012). The ecosystem service agenda: Bridging the world of natural science and economic conservation and development and public and private policy. *Ecosystem Services* 1 : 4-15

Brussard, I, De Ruiter, P.C and Brown G. G (2007) Soil biodiversity for agricultural sustainability. *Agric . Ecosystem Enviroment* 121: 233-244

Fischer, G., F. Nachtergaele, S. Prieler, H.T. van Velthuizen, L. Verelst, D. Wiberg, 2008. Global Agro-ecological Zones Assessment for Agriculture *(GAEZ 2008)*. IIASA, Laxenburg, Austria and FAO, Rome, Italy.

Gebreyesus, B.T, Lulseged T.and Paul L.G. V. (2011). *CATENA* Volume 86, Issue 1, Pages 1-13. https://doi.org/10.1016/j.catena.2011.01.013

International Year of Soils, 2015: https://www.fao.org/soils-2015/news/news-detail/en/c/277682/

ITPS, State of Knowledge of Soil Biodiversity, 2020 Report; 1-583pp.

Karlen D.L. *et al.* (2001) **Soil quality: current concepts and applications. Advances in Agronomy**

Mikha, MM; Vigil, MF; Liebig, MA; Bowman, RA; McConkey, B; Deibert, EJ; Pikul, JL (2005). Cropping system influences on soil chemical properties and soil quality in the Great Plains. *Renewable Agriculture and Food Systems*. 21(1): 26 – 35

Pappu , H. R , J ones , R. A .C and Jain,R.K (2009). Global Status of tospovirus epidemics indiverse cropping system. Successes achieved and challenges ahead. *Virus Research* 141(2): 219-236

Sposito, G. The Chemistry of Soils. (2008) Oxford University Press - Science - 344 pages

Tsiafouli, M. A ,Thebault, E, Sgardelis S. P , De Ruiter. P. C ,VanderPuttern, W. H . Birkhofer. K (2015). Intensive Agriculture reduces soil biodiversity across Europe. *Glob. Chang. Biol* 21: 973-985.

Abel-Mawgoud, A. M. R., Sassine, Y., El- Behairy, U. A. A., Hadid, A. F. A. and El-Abd, S. O. (2005). Effect of minium root –zone temperature on the growth and production of greenhouse sweet pepper. *Journal of Applied Sciences Research*, 1 (1): 72 – 77.

Adekiya, A. O. and Ojeniyi, S. O. (2002). Evaluation of tomato growth an soil properties under methods of seedling bed preparation in an alfisol in the rainforest zone of Southwest Nigeria. *Soil and Tillage Research*, 64 (2002): 275 – 279.

Adekiya, A. O., Agbede, T. M. and Ojomo, A. O. (2009). Effects of tillage methods on soil properties, nutrient content, growth and yield of tomato in an alfisol of Southwestern Nigeria. American-Eurasian Journal of Sustainable Agriculture, 3 (3). 348 – 353.

Al-shammary, A. A. G., Kouzani, A.Z., Kaynak, A. and Khoo, S. Y. (2018). Soil bulk density estimation method: a review. *Pedosphere*, 28(4): 581 – 596.

Akinyele, B. O. and Adigun, A. B. (2006). Phentypic expression of maize (*Zea mays* L). *Research Journal of Botany*, 1: 139 – 143.

Alagba, R.A., Obiefuna, J.C., Ibeawuchi, I.I., Onyewuchi, O.P., Ofor, M.O., Okoli, N. A., Adikuru, N.C., Emma-Okafor, L.C. and Peter Onoh, C.A.. (2017). Depth of planting and mulching: Critical agronomic aspects (cultural practices) in late-season plantain establishment in rainforest agroecology of southeastern Nigeria, *FUTO Journal Series*, 3 (1): 38 – 55.

Ali, O. N. (2018). Temperature effect on maize germination and root elongation. Theses and Dissertations, 4230.

Amare, G. and Desta, B. (2021). Coloured plastic mulches: impact on soil propertitiesz and crop productivity. *Chemical and Biological Technologies in Agriculture*, 8 (4): 1 – 9.

Diatta, A. A., Thomason, W. E., Addaye, O., Vaughan, L. J., Thompson, T. L., Lo, M., Chim, B. K. and Bateman S. (2018). Innoculation and soil texture effects on yield and yield components of mungbean. Journal of Agricultural Science, 10 (9): 6 – 16.

Atiyeh, R. M., Lee, S., Edwards, C. A., Arancon, N. Q. and Metzger, J. D. (2002). The influence of humic acids derived from earthworm-processed organic wastes on plant growth. *Bioresour Technol*, 84: 7 – 14.

Aburab, M. E. El-Mogy, M. M., Hassan, A. M., Abeldaym, E. A., Abdelkader, N. H. and El-Sawy, B. I. (2019). The effects of root aeration and different soil conditioners on the nutritional values, yield and water productivity of potato in lay Loam soil. *Agronomy*, 9: 8.

Bender, S. F. and van der Heijden, Marcel G. A. (2014). Soil biota enhance agricultural sustainability by improving crop yield, nutrient uptake and reducing nitrogen leaching losses. *Journal of Applied Ecology*, 52 (1): 228 – 239.

Brady, N. C. and Weil, R. R. (2008). The nature and properties of soils, 14th edn. Pearson/Prentice Hall, Upper Saddle River.

Chaves, B., Hoogenboom, G. and Thornton, P. (2013). Variability in maize yield due to differences in soil texture and the effect of water and nitrogen. Proceedings of the ASA, CSSA and SSSA. International Annual Meetings, held in Tampa, Florida, USA, 3 – 6 November, 2013.

Cardone, L., Castononuovo, D., Perniola, M., Scrano, L., Cicco, N. and Candido, V. (2020). The influence of soil physical and chemical properties on saffron (Crocus sativus L.) growth, yield and quality. *Agronomy*, 10 (1154).

Ddamulira, G., Malaala, A., Otim, A., Florence, N. and Maphosa, M, (2022). Soil amendments improved tomato growth, yield and soil properties. *American Journal of Plant Plant Sciences*, 13: 960 – 971.

Duo, F., Soriano, J., Tabien, R. E., Chen, K. (2016). Soil texture and cultivar effects on rice (Oryza sativa L.) grain yield, yield components and water productivity in three water regimes. *PLOS ONE*, 11 (13): e0150549.

Ebubekir, A., Taser, O. F. and Kara, O. (2008). Yield of silage maize as affected by compaction treatments at the planting time. *Asian Journal of Plant Sciences*, 7: 223 – 227.

Ekpe, I.I., Oti, N. N., Okonkwo, K. C., Nwankwo, V. C., Orji, A., Okoli, N. A. , Aguwa, U. O. and Ojiako, F. O. (2021). Performance of pig slurry on a degraded sandy soil and the yield of irrigated upland rice (*Oryza sativa*) in Owerri, Southeast Nigeria, *Discovery Agriculture*, 7 (17). 26 – 36.

Emma-Okafor, L. C, Obiefuna, J. C, Ibeawuchi, I. I, Onweremmadu, E. U, Okoli, N. A. and Okere, S. E. (2016), Soil Fertility Maintenance In Late Season Plaintain Production Using Poultry Manure And Time Of Planting in Owerri, Southeastern Nigeria. *FUTO Journal Series*, 2, (1): 9 – 16

Francisco, J. C., Louise, E. J., Kate, M. S. and Dennis, E. R. (2000). Microbial response to simulated tillage in cultivated and uncultivated soils. *Soil Biology and Biochemistry*. 32 (11-12): 1547 – 1559.

Goodman, A. M. and Ennos, A. R. (1999). The effects of soil bulk density on the morphology and anchorage mechanics of the root systems of sunflower and maize. *Annals of Botany*, 83 (3): 293 – 302.

Heloisa, O. S., Renato, C. C. V, Beatriz P., Raquel, M. O.P, Elise M. P., Giovani, V. T and Edila, V. R. P. (2019). Effect of soil temperature in the emergence of maize seeds. *Journal of Agricultural Science*, 11 (1): 479 – 484.

Hussain, S., Khaliq A., Ali, B., Husaain, H. A., Qadir, T. and Hussain, S. (2019). Temperature extreme: Impact on rice growth and development. In: Hasanuzzaman, M., Hakeem, K., Nahar, K., Alharby, H. (eds) Plant abiotic stress tolerance. Springer Nature Switzerland AG.

Gholami., A., Biyari, A., Gholipoor, M. and Rahmani, H. A. (2012). Growth promotion of maize (Zea mays L.) by plant growth promoting rhizobacteria under field conditions. Communications *in Soil Science and Plant Analysis*, 43: 1263 – 1272.

Ibeawuchi, I. I, Obiefuna, J. C, Alagba, R. A, Okoli, N. A, Ofor, M. O., Emma-Okafor, L. C, Iwuanyanwu, U. P, Nze, E. O. and PeterOnoh, C. A. (2015). Organic Matter Management For Sustainable Tropical Crop Production. *Journal of Natural Science Research*, 5(1):153 – 163.

Isam, A. F. M. Saiful, Kitaya, Y., Hirai. H., Yanase, M. and Mori, G and Yiyota, M. (1998). Growth characteristics and yield of carrots grown in a soil ridge with a porous tube for soil aeration in a wet lowland. *Scientia Horticulturae*, 77 (1-2): 117 – 124.

Indrajit (2023). Soil aeration: Importance and effects. Soil Management. India

Khalil, H. P., Hossain, M. S., Rosamah, E., Azil, N. A., Saddon, N., Davoudpoura, Y., Islam, M. N. and Dungani, R. (2015). The role of soil properties and its interaction towards quality plant fiber: A review. *Renew. Sustain. Energy Rev*, 43 : 1006 – 1015.

Lamidi, W. A. Shittu, K. A. and Adeyeye, A. S. (2018). Yield responses of tomatoes (Lycoperscion esculentum) on organic manure buffered lateritic soils. *Journal of Applied Science, Environment and Management*, 22 (8): 1207 – 1212.

Leeper, G.W. and Uren, N.C.(1993). Soil Science: An Introduction, 5th edn. Melbourne University Press, Melbourne.

Ma, Z. Y., Xie, Y., Zhu, L., Cheng, L.. Xiao, X., Zhou, C. and Wang J. (2017). Which of soil microbes is in positive coorelation to yields of maize (Zea mays L)? *Plant Soil Environment*, 63: 574 – 580.

Meng, Q, Li, D, Zhang, J., Zhou, L., Ma, Z., Wang, H., and Wang, G. (2016). Soil properties and corn (*Zea mays* L.) production under manure application combined with deep tillage management in solonetzic soils of Songnen Plain, Northeast China.*Journal of Integrative Agriculture,* 15 (4): 879 – 890.

McLaren, R. G. and Cameron, K. C. (1996). Soil Science sustainable production and environmental protection, Oxford University Press: Auckland, New Zealand, pp. 340.

Merriam-Webster Dictionary (2017). Crop.

Nimmo, J. R. (2004). Porosity and pore size distribution. In: Hillel, D., Ed., Encyclopedia of soils in the environment, Elsievier, London, 295 – 303.

Olymbios, C. M. and Schwabe, W. W. (1976). Effects of aeration an soil compaction on growth of the carrot, Daucus carota L. *Journal of Horticultural Science*, 52 (4):

Okoli, N. A. and Okocha, S. K. (2022). Soil moisture deficit effect on nutrient contents of cassava leaves during the early and late cropping seasons in soils amended with poultry manure - palm bunch ash mixture. *Thai Journal of Agricultural Science,* 55 (1): 45 – 64.

Okoli, **N. A**, Obiefuna, J. C, Ibeawuchi, I. I, Ihejirika, G. O and Alagba, R. A (2011): Assessment of poultry manure fortified with palm bunch ash (POASH) for cassava production in southeastern Nigeria. *Acta Agronomica Nigeriana.* 2: 88 - 97.

Oshunsanya, S. and Nwosu N. (2018). Soil-water relationship: a case study of cassava in the tropics. Intechopen, pp. 163 – 181.

Niu, Y., Luo, Z., Cal., L., Coulter, J. A., Zhang, Y., Berti, M. Continuous monocultute of alfalfa and annual crops influence soil organic matter and microbial communities in the rainfed loess plateau of China. *Agronomy*, 10 (1054).

Pregitzer, K. and King J. (2005). Effects of soil temperature on nutrient uptake. In: BassiriRad, H. (eds) Nutrient acquisition by plants. Ecological Studies, vol 181. Springer, Berlin, Heidelberg, pp. 277 – 310.

Roberts, P., Jones, D. L. and Edward-Jones, G. (2007). Yield and vitamin C content of tomatoes grown in vermicompost wastes. *J. Sci. Food Agric,* 87: 1957 – 1963.

Ramaswwamyreddy, S., Basavaraju, P., TV, R.D. and MA, N. (2021). Effect of Water and Soil Quality on Crop Productivity. In: Leal Filho, W., Azul, A.M., Brandli, L., Lange Salvia, A., Wall, T. (eds) Clean Water and Sanitation. Encyclopedia of the UN Sustainable Development Goals. Springer, Cham.

Sabri, N. S. A., Zakaria, Z., Mohamad, S. E., Jaafar. A. B. and Hara, H. (2018). Importance of soil temperature for the growth of temperate crops under a tropical climate and functional role of soil microbial diversity. *Microbes and Environment*, 33 (2): 144 - 150.

Sustainable Agriculture Research and Education (SARE) (2021).Building soils for better crops: In Fred Magdoff and Harold van Es (eds). Ch 5. Soil particles, water and air. SARE Outreach. Pp.410.

Seiko O., Hidenori, I and Toru, K. (1994). Effect of soil aeration on the occurrence of club disease of crucifers. *Soil Science and Plant Nutrition*, 40 (3): 445 – 455.

Shreeja, D. (2023). Soil and plant growth. Soil Management, India

Silva-Neta, I. C., Pinho, E. V., Veiga, A. D., Pinho, R. G., Guimaraes, R. M., Caixeta, F., and Marques, T. L. (2015). Expression of genes related to tolerance to low temperature for maize seed germination. *Genetics and Molecular Research*, 14(1): 2674 – 2690.

Sanchez-Maranon, M. (2011). Colour indices, Relationship with soil characteristics. In: Glinski, J., Horabik, J., Lipiec, J. (eds) Encylopedia of Agrophyscis, Encyclopedia of Earth Sciences Series, Springer, Dordrecht.

Shao, Xuexin (2013). Nutrient retention in plant biomass and sediments from the salt marsh in Hangzhou Bay estuary, China.

Soothar**,** R. K., Singha, A., Soomro, S. A. and Chachar, A., Kalhoro, F. and Rahaman, F. (2021). Effect of different soil moisture regimes on plant growth and water use efficiency of Sunflower: experimental study and modeling. *Bulletin of the National Research Centre,* 45, (121): 2021.

Tao, J., Lui, X., Liang, Y., Nui, J., Xiao, Y., Gu, Y., Ma, L., Meng, D., Zhang, Y., Huang, W., Peng, D., Zhang, Y., Huang, W., Peng, D. and Yin, H. (2017). Maize growth responses to soil microbes and soil properties after fertilization with different microbes and soil properties after fertilization with different green manures. *Applied Microbiology and Biotechnology*, 101 (3): 1289 – 1299.

Tao, J., Meng, D., Qin, C., Lui, X., Liang, Y., Xiao, Y., Liu, Z., Gu, Y., and Yin, H. (2018). Integrated network analysis reveals the importance of microbial interactions for maize growth. *Applied Microbiology and Biotechnology*, 102 (8). 3805 – 3818.

Uguru, M. I. (2011). Crop production tools, techniques and practices. Fulladu Publishing Company, Nsukka, Nigeria. Pp. 69 – 74.

Yuan, L., Wenguan, N., Xiaoshu, C., Jingwel, W., Mingzhi, Z., Xiaaohui, D. and Zhenxing Z. (2019). Effect of soil aeration on root morphology and photosynthestic characteristics of potted tomatoes plants (Solanum lycopersicum) at different NACl salinity levels. *BMC Plant Biology*, 19: 331.

Yumiko, A., Tsutomu, I., Akihiro, O. and Motohiko, K. (2010). Effects of soil temperature on growth and root function in rice. *Plant Production Science*, 13 (3): 235 – 242.

Zhang, Y., Kendy, E., Qiang, Y., Changming, L., Yanjun, S., Hongyong. S. (2004). Effect of soil water deficit on evapotranspiration, crop yield and water use efficiency in the North China Plain. *Agricultural Water Management*, 64 (2): 107 – 122.

Zhuo, L., Liu, J. H., Zhao, B. P., Xue, A. and Hao, G. C. (2016). Effects of soil amendment on soil charactertics and maize yield in Horqin Sandy Land. IOP Conf.Ser.: *Earth Environ. Sci*., 41: 0122005.

Zucco, M. A., Walters, S. A., Chong, S. K, Klubek, B. P. and Masabni, J. G. (2015). Effect of soil type and vermicompost applications on tomato growth. *International Journal of Recycling of Organic Waste in Agriculture*, 4 (2015): 135 – 141.

Zuo, C. B., Penfold C. Sands, R. Misra, R. K. and Hudson, I. (2001). Effects of soil-filled porosity, soil matric potential and soil strength on primary root growth of radiate pine seedlings. *Plant and Soil*, 235 (1): 105 – 115.

View publication stats